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Design of Conversational Experiences

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Lecture No. 7

Agenda



- Monetization
- Case Studies of PTOBot and VacationBot



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Monetization

Chapter -13, 15,16 at Textbook T1

T1: Designing Bots. Creating Conversational Experiences by Amir Shevat

Monetization



Two types of monetization

1. Bot Builder like RASA, ChatFuel using which we build bots
2. A chatbot which is deployed on a Web / FB platform, where users pay for the services that the bot offers

KEY TAKEAWAY

End users do not pay for bots, they pay for the services the bots expose.

Subscription



- This is currently the most common way that bots drive revenue – the bot provides an ongoing service that the user subscribes to and pays for.

Upgrade Growbot to fuel insights and promote your company values

Growbot Premium includes access to premium dashboards, employee leaderboards, company values support, and more.

Basic	Premium
\$0	Contact Us
For small teams & evaluations	Billed monthly
3 keywords	Unlimited keywords
No customization	Custom keywords
Your current plan	Dashboards
	Leaderboards
	Company values
	I'm interested in Premium

Figure 13-2. Growbot experimenting with premium services

Subscription Categories



1. Basic features - free, no payment required
2. Premium features - need to pay

Accelerator	Growth	Enterprise
Send WhatsApp messages in bulk with WhatsApp API and manage conversations with the WhatsApp inbox. Start for free ₹ 7000/ qtr ✓ Green tick assistance ✓ Broadcast & Scheduling ✓ WhatsApp Inbox ✓ Basic Automation (Welcome, OOO) ✓ GSheets integration ✓ Add upto 5 agents ✓ Analytics for Broadcasts & Livechat ✓ Message templates catalog ✓ Retargeting Campaigns ✓ Click to WhatsApp Ads (CTWA)	Grow your WhatsApp marketing and run smart campaigns with advanced automation & analytics. Start for free Book a Demo ✓ Everything from Accelerator + ✓ Advanced chatbot flows ✓ Custom campaign flows ✓ CRM integration ✓ Access to WhatsApp flows ✓ Advanced Analytics with CTWA, Chatbot, and Campaigns ✓ Run intent based campaigns ✓ FAQs ✓ 1000 MAUs	Build custom WhatsApp plan most suited for your business needs Start for free Book a Demo ✓ Everything from Growth + ✓ eSenseGPT - GenAI powered conversations ✓ Multiple digital channels ✓ Omnichannel Livechat Inbox ✓ Custom app integrations ✓ Unlimited chatbot flows ✓ Full-view Analytics ✓ LLM integrations ✓ Voice integration

Subscription



- Note that even though the payment itself is done on the web, it is important to acknowledge it in the chat interface.
- Notifying the user of upgrades, downgrades, and recurring payments is a good way to promote transparency and user satisfaction.
- The subscription model, when done right, is very **lucrative** because the lifetime value of each user (the amount of money the user will pay in the lifetime of using your product) is usually higher than with other models.

Monetization in Bot



Bots can be monetized through:

1. Ad Serving
2. Data Collection from user
3. Data Analytics
4. Market Research
5. Selling Goods and Services
6. Referral Fees
7. Brand Promotion
8. In-Bot Virtual Goods

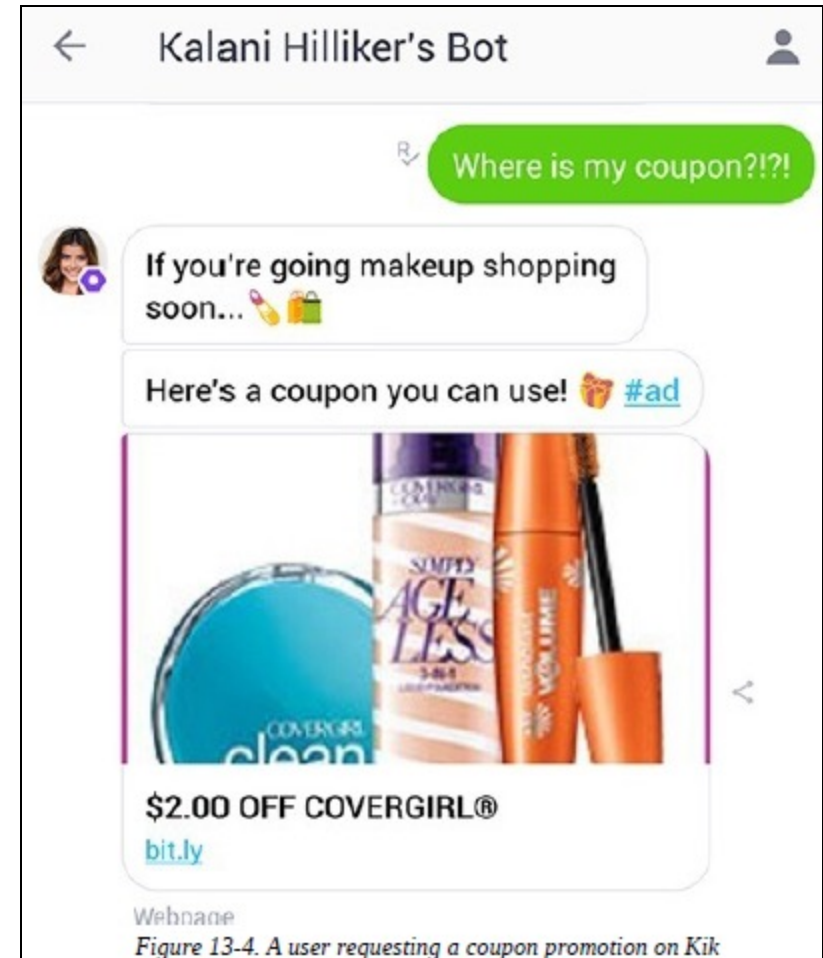
1. Ad Serving



- Primarily a major means of earning for the Web and Mobile interfaces

There are two ways:

1. User asks the bot for the ad (coupon), which serves as an “intent” to display the ad
2. During the conversation, bots display the “ads” to the user, as a means of selling



Webname

Figure 13-4. A user requesting a coupon promotion on Kik

- Bots can collect a lot of data about user preferences and interests, through engaging in a conversation or playing a game.

Primary User Data

- Primary user data (first-party data) is any information the person willingly shares with a brand via emails, social media, live chat, or chatbots.
- Such data may include the user's email, name, shoe size, color preferences, or feedback regarding your service.
- May be collected via Forms

Data Collection Strategies



- User Input : form of word, sentence, documents etc.
- Entity : to collect user name, email etc.

Post data collection

- Segments: form users into segments, to send better notifications
- Attachments: Analyze uploaded documents
- Export chat data: integrate with google sheets, Salesforce, Shopify orders etc.
- Generate sales lead

Ref: <https://www.chatbot.com/academy/chatbot-designer-free-course/data-collection/>

Market Research



- Now imagine you were a fast food brand, for example, and wanted to know which new type of sandwich customers would prefer, or what type of fast food imagery they would find more appealing.
- This type of bot would be a gold mine for your market research.
- The Swelly bot can generate detailed reports on audience preferences in a precise and very fast way. While this bot does not charge the end user, it will make a lot of money helping brands understand their audience.

Try:

Swelly.ai (FB Messenger) -> <https://www.swelly.ai/>

Selling Goods and Services



- Bots can also become the channel through which you sell goods and services.
- These can be tangible goods that the bot sells directly in chat, or paid services that the bot exposes through chat.
- Ex: Ride services Lyft and Uber, who have interfaces in Slack and FB channels



Figure 13-6. The Lyft bot on Slack

Referral Fees



- This is another major business model on mobile and web that is moving to bots - the bot can help you decide what to buy or which service to consume and then refer you to the right service, rather than actually completing the transaction itself.

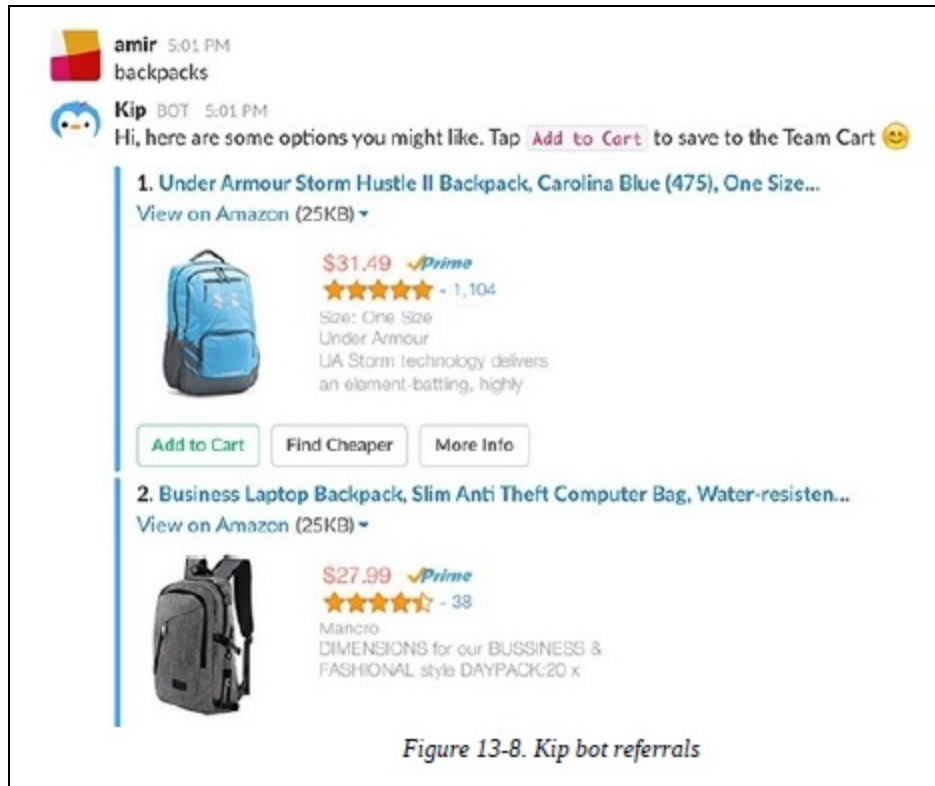


Figure 13-8. Kip bot referrals

Brand Promotion



- Another indirect way to drive your business through a bot is by promoting brand recognition. Thinking about the bot as a frontend representative of your product or service can make a lot of sense, and having a delightful bot that provides a useful service can generate strong attachment to your brand.



Figure 13-9. The H&M bot on Kik

In-Bot Virtual Goods



- Sale of in-bot virtual goods is another model that is going to be big in upcoming years.
- An example would be a game bot that sells you chips, or a bot that sells you music and photos.
- One of the most lucrative models in the mobile app industry.

When to start charging users?



- You should only start considering charging for your bot when you feel that you have reached product/market fit.
- Users need to get hooked on your product, use it on a regular basis, and derive a lot of value from it before they are asked to pay for the bot.
- Whether you are providing a trial, a freemium version, or in-bot virtual goods, make sure your users love your product before you try to charge them.

KEY TAKEAWAY

You should only start considering charging for your bot when you feel that you have reached product/market fit.



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Case Studies

Case Studies



PTOBot

- The PTO (paid time off) bot will help an employee secure his/her manager's approval to take a vacation.

VacationBot

- The vacation bot will help an employee find fun things to do while on vacation.

PTOBot Requirements



- PTOBot will enable users to perform actions in the PTO-IT system
- PTOBot will enable the user to request PTO, setting dates and providing a description
- PTOBot will enable managers to approve PTO requests
- PTOBot will record the approval/rejection status of each request and communicate it to the employee.
- PTOBot will enable an employee's teammates to get notifications when their colleague is taking a vacation.
- PTOBot will promote VacationBot to the user at the end of the approval process.

PTOBot Persona



Name: PTOBot

Environment: Work

Audience: Adults aged ~25-55, early adopters

Task at hand: PTO requests

Runtime variations: Report abuse, zero tolerance to not safe for work (NSFW) content

Locally relevant social acceptance: Work environment

Service branding: Professional, productive

Values: Getting things done, and fast

Derived personality: Serious, to the point, friendly but not humoristic, safe for work. This bot should be like a company office manager or a personal assistant.

VacationBot Requirements



- VacationBot will only be distributed by PTOBot and work for PTO-IT clients
- VacationBot will get a parameter in the referral from PTOBot, and will be able to retrieve vacation start and end dates from PTO-IT.
- VacationBot will support sending notifications to the user, as well as providing the ability to get a list of events and activities at the vacation destination on demand.
- VacationBot will promote subscribing to notifications about events and activities to the user during the vacation.
- VacationBot will enable linking from each notification to the relevant event or activity with the appropriate affiliation link.

VacationBot Persona



Name: VacationBot

Environment: Consumer

Audience: Adults aged ~25-55, early adopters

Task at hand: Provide fun activity recommendations

Runtime variations: Provide a more casual conversation during the vacation

Locally relevant social acceptance: Family-friendly

Service branding: Professional, fun but still not too casual

Values: Enriching employees lives while they're on vacation

Derived personality: Friendly but not too humoristic, family-friendly, safe for work. This bot should be like the concierge service in a high-end hotel.



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Conversation Scripting for PTOBot and VacationBot Chapter16 :T1

High Level Conversation Flow

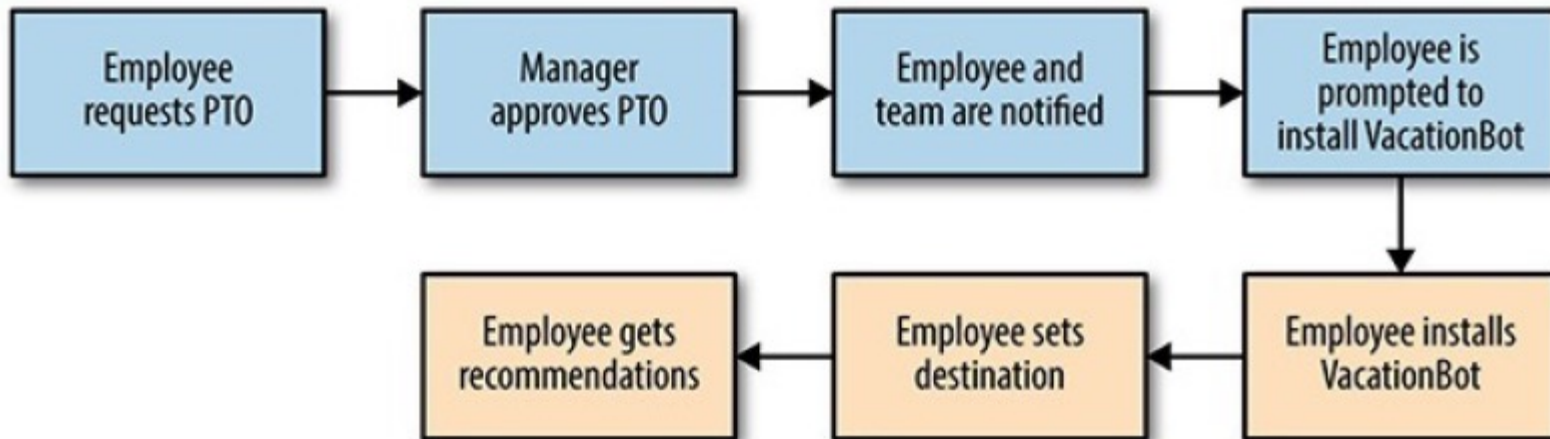


Figure 16-1. A high-level view of the PTOBot/VacationBot flow

PTOBot Onboarding Flow

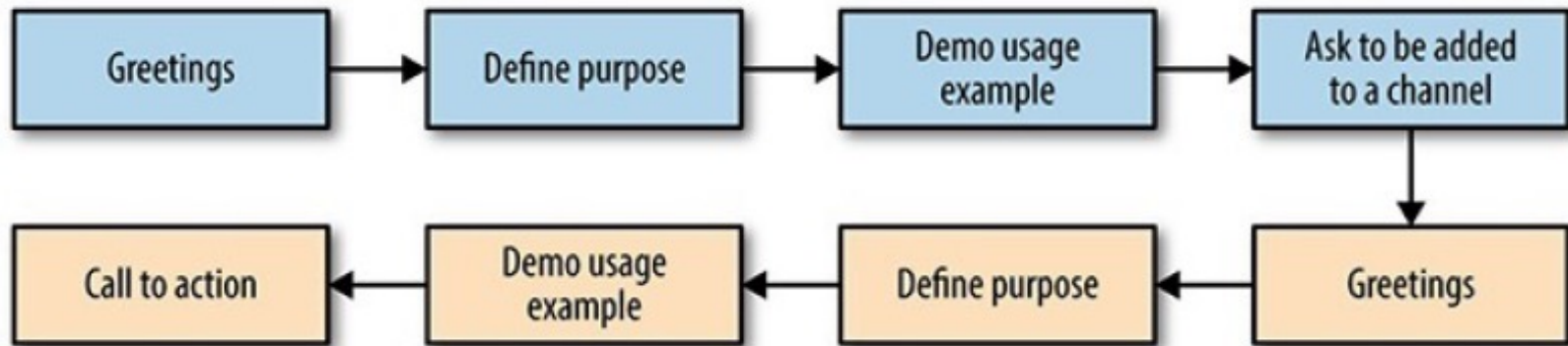


Figure 16-2. PTOBot onboarding flow

PTOBot Main Flow

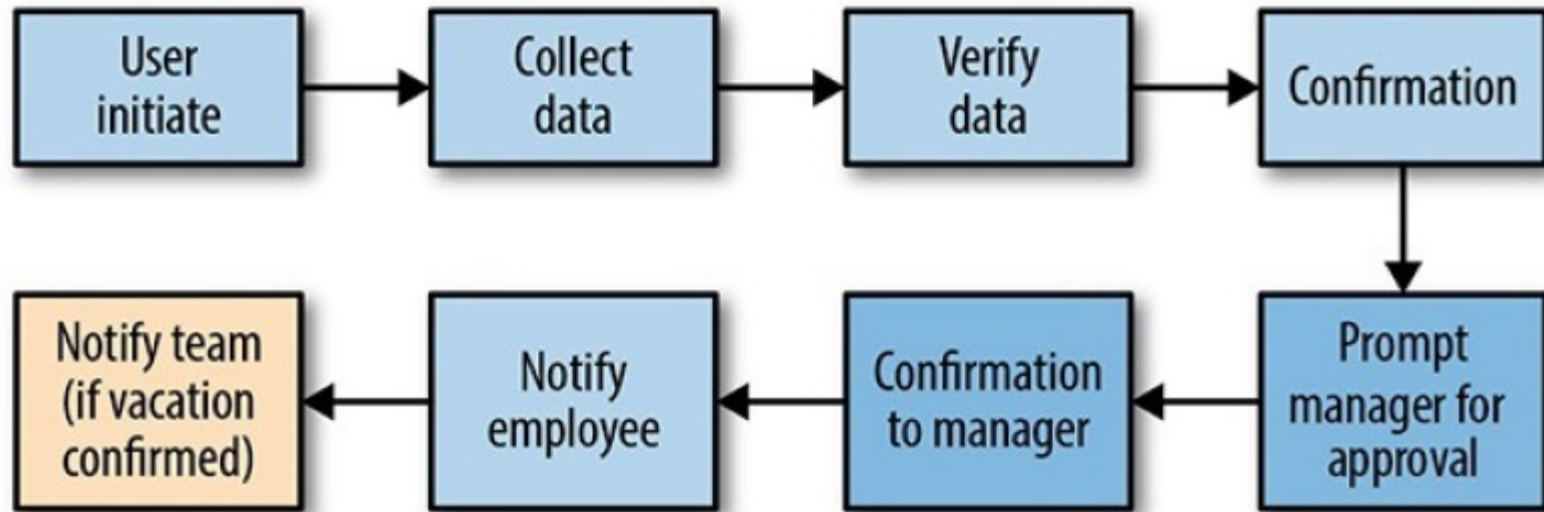


Figure 16-4. PTOBot main flow

Vacation Bot Main Flow

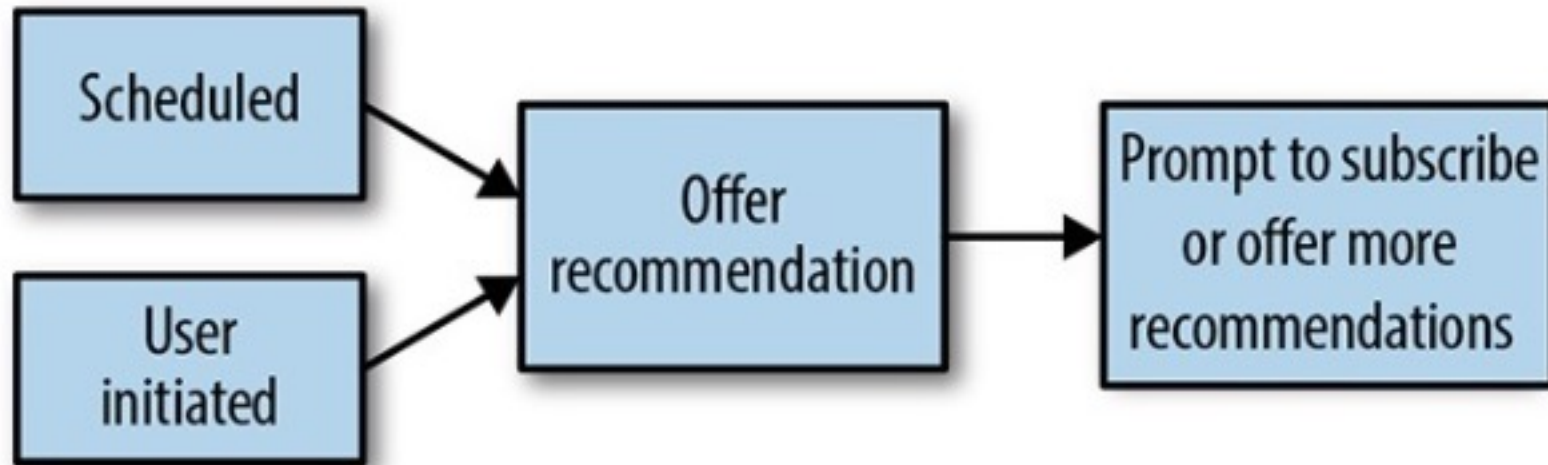


Figure 16-5. VacationBot main flow

PTOBot Help Flow

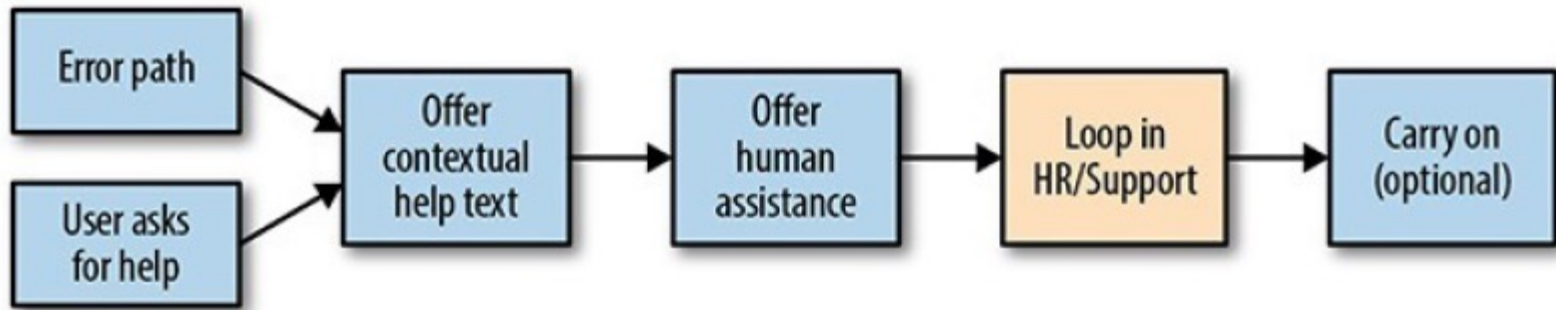


Figure 16-6. PTOBot help flow

VacationBot Onboarding Flow

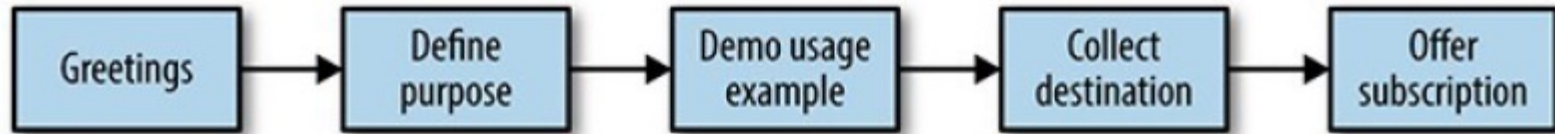


Figure 16-3. VacationBot onboarding flow

VacationBot Main Flow

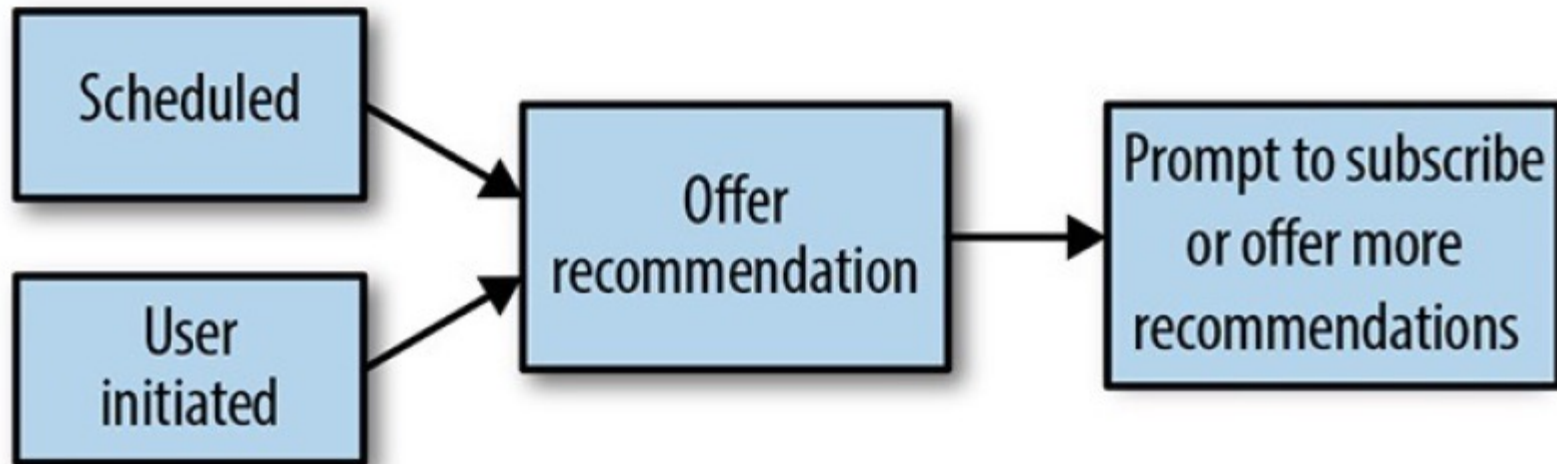


Figure 16-5. VacationBot main flow

VacationBot Help Flow

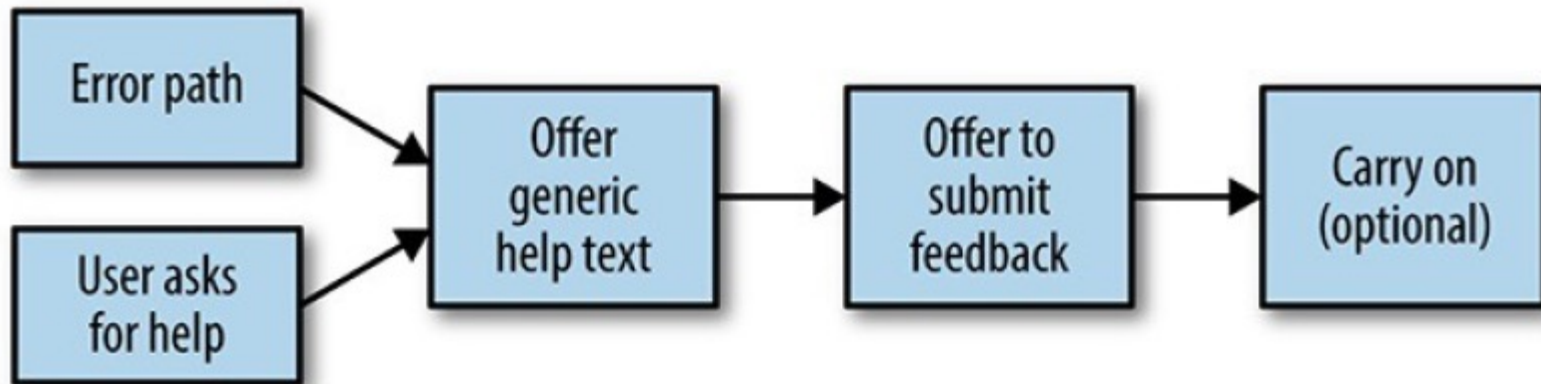


Figure 16-7. VacationBot help flow

Feedback Flow (Same for both Bots)

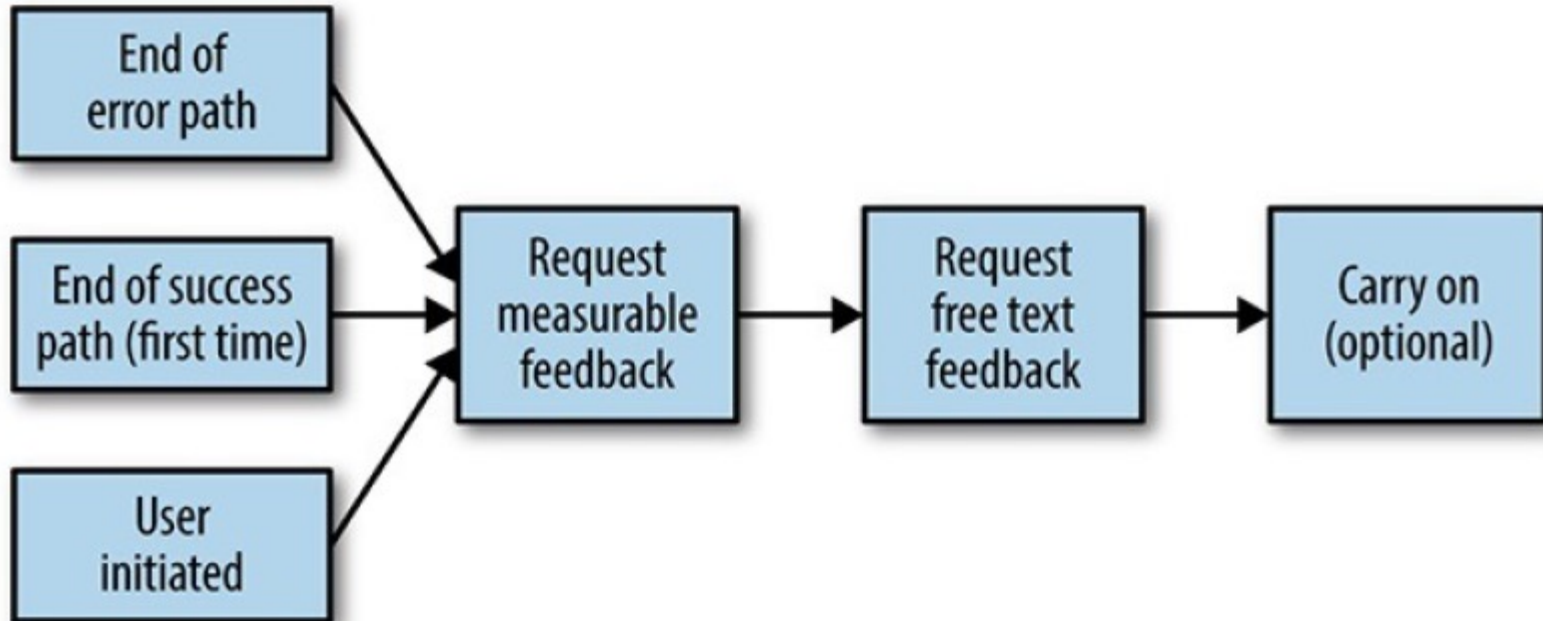


Figure 16-8. Feedback flow

Error Flow (Same for both Bots)

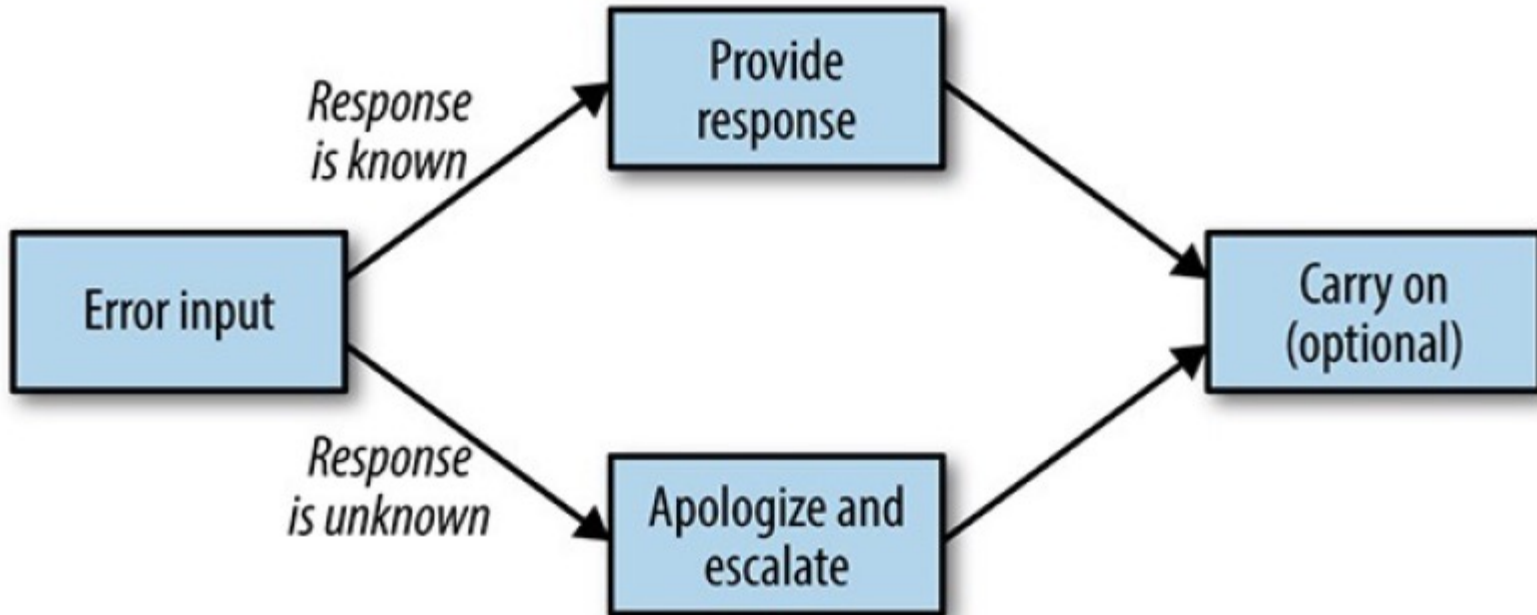


Figure 16-9. Error flow

Intents and Sample Utterances



For PTOBot

1. **PTO request:** "PTO", "Hi", "Hello", "Start", "Vacation"
2. **Help:** "Support", "Help", "Not sure", "What?"
3. **Feedback:** "This is great", "This sucks", "Feedback"

For VacationBot

1. **Get activities:** "Recommendations", "Hi", "Hello", "What's up", "News"
2. **Help:** "Support", "Help", "Not sure", "What?"
3. **Feedback:** "This is great", "This sucks", "Feedback"

Entities Identification



For PTOBot, the PTO request entities are:

- Start date {Date}
- End date {Date}
- Description {Text – up to one paragraph}
- Approved {Yes/No}

For VacationBot, the vacation activity entities are:

- Destination {Location}
- Schedule notifications {Yes/No}
- When to send the notification {Morning/Evening}

Onboarding Conversation for PTOBot



PTOBot

The onboarding script for our PTOBot is split into two separate conversations: the 1:1 one with the user who is installing the bot, and a team conversation.

The conversation with the installer might look like this:

PTOBot: Thank you for adding me to the team! I am an HR bot provided to you as part of your PTO-IT solution. I can facilitate PTO requests and approvals.

PTOBot: Here is how I work: [show a GIF of working with the bot, one image with an employee and the other with a manager]

PTOBot: In order for me to work well, please add me to the relevant channels - recommended channels are #HR and #general.

The conversation with the team (once the bot has been invited to the channel) might go like this:

PTOBot: Hi everyone, I am PTOBot, your new HR bot added by [Installer name]. I can facilitate Paid Time Off (PTO) requests and approvals.

PTOBot: Here is how I work: [show a GIF of working with the bot, one image with an employee and the other with a manager]

PTOBot: You can Direct Message me at any time to start a PTO request. You can find me at @ptobot.

PTOBot: Another way to start a PTO request is with a slash command.

PTOBot: Just type: `pto-it [start-date MMDD/YY] [end-date MM/DD/YY] [description]`

Onboarding Conversation for VacationBot



VacationBot: Thank you for connecting!

VacationBot: I am your friendly VacationBot. I can provide you with news about activities happening while you are on vacation. I will even include your corporate discounts, provided to you by PTO-IT!

VacationBot: Our records show that you are heading out on 04/21/17.

VacationBot: Please provide your destination so we can get started.

User: Cancun Mexico

VacationBot: Fantastic place for a vacation! Here are a few cool attractions in the area:

Onboarding Conversation for VacationBot



VacationBot: [Carousel with 3 activities]

VacationBot: I can send you fun new activities every day. Would you like that?

VacationBot: [Yes, every morning] [Yes, every evening] [No thanks] (*Quick Replies*)

{ Yes, every evening }

VacationBot: Fantastic, will do. Have a great vacation! You can always say “Recommendations” to get more recommendations at any time.

{No}

VacationBot: That is fine, you can always say “Recommendations” to get more recommendations at any time. Have a great vacation!

VacationBot: [Recommendations] (*Quick Reply*)

PTOBot Main Flow



Four Scripts

1. PTO Request
2. PTO Approval
3. PTO Approval Notification
4. Team Notification

PTOBot Main Flow - PTO Request



User: Hi @PTOBot I would like to take a PTO

PTOBot: Happy to assist! Which date would you like to start your PTO?

User: 04/21

PTOBot: Thanks, which date would you like to end your PTO?

User: 04/31

PTOBot: Please provide a description for your PTO request.

User: Holiday vacation

PTOBot: Great! To confirm, you would like to take PTO between the 04/21/2017 and 04/31/2017 for a Holiday vacation.

PTOBot: [Make Request] [Edit] (*buttons*)

{“Make Request” button clicked}

PTOBot: Thank you! I will now send the request to your manager, April Underwood, to approve your PTO.

User: Thanks!

PTOBot: You are welcome.

PTOBot Main Flow - PTO Approval



Here's our example manager PTO approval script (between the manager and PTOBot in a direct message):

PTOBot: Hi April, Chris Messina would like to take PTO between **04/21/2017** and **04/31/2017**. If approved, Chris will have **5 days in surplus** remaining.

PTOBot: [Approve] [Reject] (*buttons*)
{“Approve” button clicked}

PTOBot: Thank you, I will notify Chris Messina that the PTO request has been approved.

PTOBot Main Flow - PTO Approval Notification



PTOBot: Good news! Your manager, April Underwood, has approved your PTO request: *(formatted in a message attachment)*

PTOBot: Would you like me to send a notification in the #PTO channel?

PTOBot: [Notify Team] [Skip] *(buttons)*

{Any button clicked}

PTOBot: If this PTO is a vacation, you might be interested in our new VacationBot offer. This is a Facebook Messenger bot that can inform you about activities and corporate discounts at your destination.

PTOBot: Here is the link to install it - *[link to vacation bot]*

PTOBot Main Flow – Team Notification



PTOBot: PTO alert:

PTOBot: Employee: Chris Messina

PTOBot: When: 04/21/17 04/31/17

PTOBot: Approved by April Underwood

PTOBot Help Script



PTOBot: Here is how I work: [show two GIFs of working with the bot, one with an employee and the other with a manager]

PTOBot: You can direct message me at any time to start a PTO request. You can find me at @ptobot.

PTOBot: Another way to start a PTO request is with a slash command.

PTOBot: Just type: *pto-it* [start date in MMDD/YY format] [end date in MM/DD/YY format] [description]

PTOBot: [Need human help] (*button*)

PTOBot Help Script



PTOBot: Here is how I work: [show two GIFs of working with the bot, one with an employee and the other with a manager]

PTOBot: You can direct message me at any time to start a PTO request. You can find me at @ptobot.

PTOBot: Another way to start a PTO request is with a slash command.

PTOBot: Just type: *pto-it* [start date in MMDD/YY format] [end date in MM/DD/YY format] [description]

PTOBot: [Need human help] (*button*)

PTOBot Feedback Script



Bot: We would love to get your feedback!

Bot: How would you rank your experience:

Bot: [Great] [Good] [Poor] [Terrible] (*buttons*)

{Any button clicked}

Bot: Please provide your verbal feedback in up to one paragraph.

User: Love this bot, I really want to take it to every vacation!

Bot: Thank you, your feedback was submitted.

Vacation Bot implementation using RASA

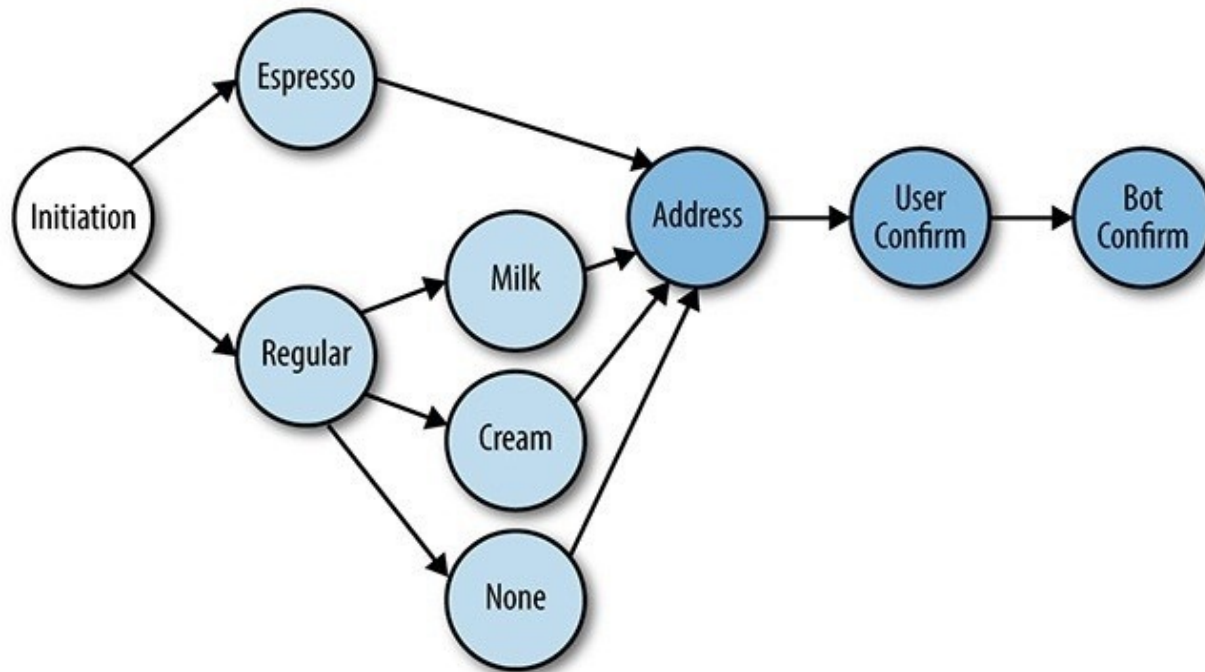


- Identify and design Intents, Utterances, Entities, Slots etc.
 - `rasa train` [to build the model]
 - `rasa run actions` [to run actions.py] in separate terminal
 - `rasa shell` [to test the bot] OR `rasa interactive`

Exercise III [15 minutes]



Design the conversation flow for “Coffee Bot” using RASA open source tool





Revision for Mid-Sem Exam (Self Study)

[Course Material uploaded to MS Teams]



Thank You!